Master Test Bench Procedures

Procedure 3.1: Emergency Power-Off (EPO) Hardwired Interlock Latency (SKY-REQ-011)

- Objective: Measure the millisecond response time from EPO switch depression to $480VAC$ high-voltage grid disconnection.
- Traceability: Requirements SKY-REQ-011, SKY-REQ-009; Interface ICD-IF-006; Hazard HAZ-005.
- Test Bench & Instrumentation:
    1. Tektronix 4-Channel Digital Storage Oscilloscope (1 GS/s sampling rate).
    2. Fluke 87V Digital Multimeter (DMM).
    3. $24VDC$ Hardwired Normally Closed (NC) EPO Switch Injector Panel.
    4. $480VAC$ Main Contactor Breakout Enclosure.
- Step-by-Step Execution:
    1. Wire Channel 1 of the oscilloscope across the $24VDC$ EPO switch dry contacts.
    2. Wire Channel 2 across the coil terminals of the $480VAC$ primary grid contactors.
    3. Configure the oscilloscope trigger mode to Single-Sweep Falling Edge on Channel 1 at a $12.0VDC$ threshold.
    4. Energize the $24VDC$ control bus and confirm the $480VAC$ contactors are latched closed.
    5. Depress the manual EPO mushroom switch on the GCS control panel.
    6. Using the oscilloscope cursors, measure the time delta (Δt) between Channel 1 voltage drop (switch actuation) and Channel 2 coil voltage decay below 10% nominal.
- Pass/Fail Threshold: $\Delta t < 20.0ms$. *(Observed baseline:* $12.4ms$ *— PASS).*

Procedure 3.2: $480VAC$ **Primary Grid Power Isolation Interlock (SKY-REQ-010)**

- Objective: Verify that primary grid power is completely isolated from the fast-charging gantry prior to mechanical connector unlatching.
- Traceability: Requirement SKY-REQ-010; Interfaces ICD-IF-001, ICD-IF-006; Hazard HAZ-004.
- Test Bench & Instrumentation:
    1. Current transducer probes attached to 3-phase $480VAC$ supply lines.
    2. PLC signal generator outputting automated connector unlatch commands.
    3. Digital oscilloscope capturing current decay profiles.
- Step-by-Step Execution:
    1. Energize the $480VAC$ primary bus feeding the PDU under a simulated $100A$ test load.
    2. Inject an unlatch request signal from the PLC controller to initiate docking disengagement.
    3. Record the current transducer output on the oscilloscope to track arc-suppression timing.
    4. Measure the time delay from the unlatch command edge until phase currents drop to zero ($0.0A$) before mechanical pin movement occurs.
- Pass/Fail Threshold: Complete current isolation in $\leq 50.0ms$. *(Observed baseline:* $34.2ms$ *— PASS).*

Master Test Bench Procedures

Procedure 3.3: Wireless Telemetry Round-Trip Time (RTT) Latency (SKY-REQ-012)

- Objective: Measure bidirectional data latency between incoming aircraft avionics and the supervisory Ground Control Station (GCS).
- Traceability: Requirement SKY-REQ-012; Interface ICD-IF-004.
- Test Bench & Instrumentation:
    1. Spirent C50 Network Traffic Generator / Wireshark packet analyzer.
    2. Private $5G$ NR base station emulator / IEEE 802.11ax wireless bridge.
    3. eVTOL Flight Control Computer (FCC) telemetry hardware simulator.
- Step-by-Step Execution:
    1. Establish an encrypted Ethernet/IP session over the private wireless network between the FCC emulator and the GCS server.
    2. Transmit $10,000$ UDP telemetry bursts ($1024 bytes$ per packet containing battery SOC, GPS, and thermal data) at a rate of $100 Hz$.
    3. Hardware-timestamp packet departure (T_{sent}) and response acknowledgment (T_{ack}) at the network interface layer.
    4. Calculate $RTT = T_{ack} - T_{sent}$ for all $10,000$ iterations and plot the 99th percentile latency.
- Pass/Fail Threshold: $99th percentile RTT < 100.0 ms$ with 0.0% packet loss. *(Observed baseline: $42.8 ms$ — PASS).*

Procedure 3.4: AOI Drone Surface Defect Resolution (SKY-REQ-007, SKY-REQ-008)

- Objective: Validate post-flight optical scanning flaw detection resolution and execution duration.
- Traceability: Requirements SKY-REQ-007, SKY-REQ-008; Interface ICD-IF-007.
- Test Bench & Instrumentation:
    1. Calibrated carbon-fiber composite wing section containing artificial $0.5 mm$ surface micro-fractures.
    2. AOI inspection drone equipped with $4K$ optical camera and LIDAR payload.
- Step-by-Step Execution:
    1. Position the composite test panel on the TLOF landing pad.
    2. Command the AOI drone to launch and execute its automated point-cloud scanning flight path.
    3. Start a high-precision timer at drone takeoff and stop the timer upon scan upload completion to the GCS.
    4. Run automated computer vision edge-detection algorithms on the uploaded $4K$ optical feed to locate surface flaws.
- Pass/Fail Threshold: Scan duration $\leq 90.0 s$ and 100% detection of $0.5 mm$ defect targets. *(Observed baseline: $68.5 s$ scan, $0.5 mm$ flaws identified — PASS).*